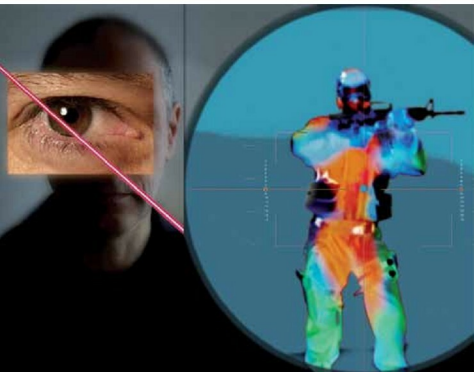


USING TECH EYES TO SPOT TERRORISTS

The terror attacks in Mumbai have highlighted the need for better surveillance in our cities. Technology could play a major role here



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The terrorist strikes in Mumbai have exposed the vulnerability of our cities. It is essential that areas like hotels, train stations, bus stations, airports, markets and malls be provided with better surveillance to enable authorities to monitor and catch suspicious characters.

The Chattrapati Shivaji Terminus (CST) at Mumbai has over three million passengers passing through it every day. Most of these people, almost all of the time, perform normal activities like buying tickets, boarding

and getting off trains, buying magazines, talking on mobile phones and so on. Statistically, an abnormal event like the terrorist attack happens once in many millions of activities. But that does not lessen its impact or the need to spot it and nip it in the bud because, although rare, such an event can cause immeasurable damage to life and property.

Is it possible to add more 'eyes' to the city? Is it possible to use closed-circuit television (CCTV) and other imaging technologies to catch terrorists before they strike? Is it possible to use machine vision technology

to automatically detect abnormal activities in the images that have been captured and alert the police? The answer to all these questions is 'yes', provided the right technology is in place.

An imaging and storage challenge

Each public place has its peculiarities. In some, it is easy to place CCTV equipment, especially indoors, where the cameras can be mounted on the walls. In others, like open markets and streets, it may be necessary to erect structures for placing the cameras.

Of course, this might need a large number of cameras. For instance, a place like CST has 18 platforms and many common areas. Covering each platform will require at least 30 cameras, and the common areas will require about 250 cameras. So, in all, CST will require about 800 cameras to cover it fully.

In open spaces like markets, tethered helium balloons, floating in the air, fitted with cameras can be used. One such balloon at a height of a few hundred metres fitted with multiple high-resolution cameras can cover the entire market very easily.

It is important to make sure that cameras capture good quality images in different conditions like daylight, night light, cloud-covered sky or heavy rain. The need is for an imaging technology that has good resolutions under different imaging conditions, and is inexpensive and robust enough to be used outdoors and indoors.

Then there is the matter of storage.